

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019**An efficacious intellectual framework for host based intrusion
detection system**Selvamani. D^{a*}, Selvi. V^b^a SIVET COLLEGE, GOWRIVAKKAM, CHENNAI - 600073, INDIA^b MOTHER TERESA WOMEN'S UNIVERSITY, KODAIKANAL- 624101, INDIA.

Abstract

Intrusion detection systems (IDS) are indispensable for all types of networks to protect them from the intruders. The IDS have to process millions of network packets with many features, which delay the detection of anomalies. To understand the attacks of the past and to design proper tools to defeat their impending peril, Host-based IDS or HIDS, establishes to be a robust design. An Efficacious Intellectual Framework for Host based Intrusion Detection with Rule Structure Generation and Pattern Matching algorithm sets the rule structure for the unknown attack generated by using Association Rule Mining in the Map Reduce Framework. It is accomplished in two stages. In the first stage, an efficacious Rule structure is generated using Intellectual method. In the second stage, Brute Force algorithm pattern matching algorithm is utilized in the proposed framework. It is proper to audit and review logs for malicious activity. Notably intrusions are rare after proven analysis to prevent the attacks to happen. The present HIDS establishes to be a robust design for understanding of attacks of the past and determining effective methods to defeat their future threats.

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Keywords: Data Mining; MapReduce; Network based Intrusion Detection System; Association Rule Mining; Brute Force algorithm

1. Introduction**1.1 Vulnerable Network**

Network security is always emphasized to ensure the secure nature of complete network in the context of increasing vulnerability of the networks. The magnitude and mode of cyber attacks have drastically changed in the recent past, as a resort for financial gain, piracy of knowledge, and political plan all over the world. The immediate need is to develop an integrated methodology that confine novel and practical solutions for many experimentation challenge to prevent possible vulnerability in the networks. The target of any hacker is this channel, with an aim to acquire the details, decode the data and reinsert a matching or identical message. Hence securing the network is just as important as securing the computers and encrypting the message. In development of a secured network, the priority lies in confidentiality, that the unwarranted party does not scrutinize the detail and integrity and that ensures the data when received is unchanged or modified after it is sent by the sender.

* Corresponding author. Tel.: +91-9940623817

E-mail address: selvamani.bhaskar@gmail.com

1.2 Intrusion Detection Systems

Intrusion Detection Systems take corrective action on a network to prevent attacks once identified in real time. A thorough vigil on the computer systems against the unauthorized use is the main goal of IDS. The IDS is a software application to detect unauthorized intrusion in the networks and computer systems which scrutinizes for unlawful activities or rule violations in a network and reports the malicious activities to the system management. An IDS monitored network makes the personnel responsible for the security of the network highly cautious of any unauthorised packets approaching or exiting the network. The Intrusion Detection in a system comprises of two types: the IDS which is host-based (HIDS) and the IDS which is network based (NIDS). The models of ID is based on three mechanisms: anomaly-based which is also based on the behaviour, signature-based ID and hybrid detection. The detection based on anomaly, focuses on detection of abnormal behaviour which is inconsistent with normal or behaviour that is anomalous. Theoretically abnormality detection requires a scheme of normal behaviour patterns. In many environments it is not practicable. Behaviour based IDs, therefore, are also considered as rule-based detection because they follow rules, developed customarily by the authority, to determine inadmissible behaviour. The signature of familiar activities is the focus of IDs based on misuse or signature-based.

1.3 Common Data mining techniques and algorithms

Wua *et al.* [1] used SVM and classification tree to compare accuracy, detection rates and false alarm rate. The result show that C4.5 is superior to SVM in accuracy and detection but in false alarm rate SVM is better. Wang *et al.* [2] have proposed a novel ID approach FC-ANN adopting fuzzy clustering and ANN. This model gives effectiveness result for R2L and U2R attacks in terms of accuracy. Canedo *et al.* [3] proposed a new KDD winner method consisting of discretizations, filters and various classifiers like Naive Bayes (NB) and C4.5 to develop robust IDS. The proposed classifier gives high accuracy i.e. 99.45% compare to others. Elbasionary *et al.* [4] have suggested hybrid technique with combination of random forest with k-mean algorithm. Baig *et al.* [5] used supervised neural network and proposed network intrusion detection model GMDH yields high attack detection rule nearly 98%. Nidhi *et al.* [6] present the neural network integrated layered framework to design an efficacious and effective IDS. The KDD dataset (Knowledge Discovery and Data Mining 1999 dataset) is used to experiment such system. Fengli *et al.* [7] proposed an effective feature selection methodology based on Bayesian Network classifier and with similar intrusion detection benchmark dataset (NSL-KDD). The performance calculation of the proposed method is carried out and measured with other feature selection methods which are normally used and shown by verifiable results. The features chosen by this method have significantly reduced the time to identify attacks and increased the classification accuracy and also the true positive rates. Fang *et al.* [8] presented a simpler intrusion detection approach that was comprehensively successful and effective by utilizing feature selection. In this both IG and ChiSquare approaches are brought together in order to decrease the chance that a lone feature selection methodology would result in biased results. Chandrasekhar and Raghuvier [9] proposed a technique consisting of four stages. The initial step, different training subset is generated with k-means and then training of many neuro-fuzzy models is achieved based upon the subset acquired. The result of KDD dataset is confirmed to demonstrate the practical application and efficiency of the evolved method, in which it shows that the proposed methods manifest desirable yield which are better than multiclass SVM, BP and further approaches.

1.4 Algorithms and HIDS

Present HIDS remain an effective design for understanding past attacks. Pattern matching algorithm takes more time to check the signature in the upcoming packets from the internet. The main objective is to impart association rule mining methodology for traffics come up in the in web services, like HTTP, HTTPS, and FTP traffic, in order to find out the strong threat features or DoS attack patens. Association rule mining is engaged in the investigation to handle the DoS patens and then find out the relationships within the features of DoS attacks in familiar dataset such as NSL-KDD. Chen *et al.* [10] proposed an approach which employs the association rule mining methodology for the appeared traffics of integrated web services, like HTTP, HTTPS and FTP, to find out the strong threat and attack features or DoS attack patens. For a framework which engages privacy-preserving mode for IDs of signature-based in a distributed network based on fog devices was proposed which demonstrates that the offered framework can assist maintain the confidentiality of data which is shared, decrease the cloud side workload, and offer less detection delay in real and simulated environments when compared with like approaches was made by Padillo *et al.* [11]. Rao *et al.* [12] considered that the pattern matching operation based on signature-based IDS would consume more memory and search time. It was aimed to match the pattern using MapReduce on a GPU by correlation method. According to Sohi *et al.* [13] an algorithm of new genetic programming for mining in Big Data association rules have been specifically designed to avoid a growing in the complexity of the solutions without an

improvement in their fitness function values. As proposed by Wang *et al.*, [14] the zero-day attacks adapt behavior change often during early existence phases to outdo the NIDS. An achievement of 94.355 in validation and Mathews correlation average where coefficient attains 0.8858 was proposed by Luo *et al.* [15] based on FASVFG classifier that obtains more accuracy in generalization.

2. Efficacious Framework for HIDS

Efficacious Intellectual Framework for Host based Intrusion Detection is proposed with Rule Structure Generation and Pattern Matching algorithm. The rule structure for the unknown attack is generated by using Association Rule Mining in the Map Reduce Framework. This proposal is composed of two stages. In the first stage, an efficacious Rule structure is generated using Intellectual method. In the second stage, Brute Force algorithm pattern matching algorithm is utilized in the proposed framework. (Fig.1a)

3. Intellectual Rule Generation Method

Stage 1: The first stage, that follows a Map Reduce design, is devoted to run an algorithm to obtain a set of ARs (RuleSet) for each subset in which the input dataset is divided.

Stage 2: Computation of Support of each Rule Set: The second stage, that also follows a Map Reduce scheme, performs the support computation of the ARs obtained in the previous stage considering all the instances of the dataset. This phase aims at evaluating the quality of the rules over the entire dataset because it is necessary in the next phase. Each Rule Set of the first phase has been only evaluated using the instances of the subset in which they have been trained. Therefore, in the second stage it is necessary to calculate the quality of the rules over the entire dataset with the aim at selecting the best rules that present the best performance in the original dataset with all the instances and not only in the subset trained.

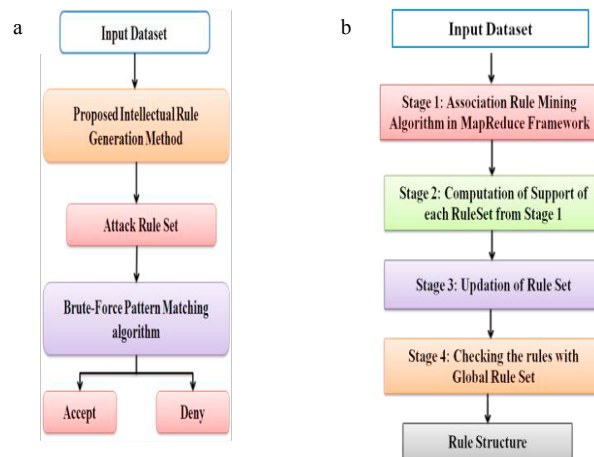


Fig.1. (a) Proposed Efficacious Intellectual Framework for Host based Intrusion Detection System; (b) Proposed Intellectual Rule Generation Method

Stage 3: The third stage sequentially updates a global rule set denominated as Global Rule Pool that stores the non-dominated solutions found in the entire dataset. To accomplish that, the same measures used by the sequential algorithm to evaluate the quality of the ARs are calculated by the support obtained in the second stage.

Stage 4: The fourth stage builds a new rule set of Global Rule Pool for unknown attacks following a Map Reduce scheme. (Fig.1b)

3.1 Construction of Intellectual Rule generation Method

The first phase, that obeys a MapReduce design, is specified to run an algorithm to derive a set of ARs (RuleSet) for each subset in which the input dataset is divided.

Stage 1: Step by Step procedure for regaining all the instances of the Map split and runs the AR algorithm in the instances of Map partition

Input: data that represents the dataset, each row is an array of values

Output: The rules discovered by the AR algorithm.

Step 1: RuleSet $\leftarrow$ {a set of ARs}

Step 2: map partitions instances $\in$ data

Step 3: {instances will contain all instances in each partition}

Step 4: association Rules $\leftarrow$ algorithm run (instances)

Step 5: end map

Stage 2: Step by Step procedure for support of each association rule computation:

3.2 Mapper Stage

Input: RuleSet a set of rules discovered by the association rule algorithm

Output: (key, value) pair, where key indicates rule Id of each QAR of RuleSet evaluated in the instance corresponding to the offset key and value contains the antecedent support (supA), consequent support (supC) and rule support(supR) of each rule in such instance.

Step 1: supports $\leftarrow$ { a key, value pair}

Step 2: map partitions instances $\in$ data

Step 3: {instances will contain all instances in each partition}

Step 4: for each Rule $\in$ RuleSet do

Step 5: {ruleId, {supA, supC, supR}} $\leftarrow$ computeSupports (instances, Rule)

Step 6: end for

Step 7: end map

The second phase is also followed in line with the MapReduce and all the instances of dataset are considered while performing the computation support of ARs of antecedent phase. Rule quality evaluation of the entire data set is the primary aim of this phase as it is necessitated in the following phase. The instances of the trained subset has been used to evaluate each of the first phase RuleSet and the same is noteworthy. Rule calculation in regard to quality over the complete dataset, aiming the selection of best performing rules in the dataset for not only in the trained subset but for all the instances.

Current association rule algorithm is redesigned to handle many data problems. Efficient objectives evaluation is performed as a result and the obtained / rule quality is improved simultaneously. To accomplish that, an intellectual rule generation method has been proposed followed by a MapReduce framework. The challenge is to discover quality association rules that represent the complete dataset through subset of rules obtained in different splits of the dataset. The association rule algorithm is only executed in a subset (Map), thus, the fitness function evaluates only a subset of instances. The evolutionary process for each Map ends when a number of evaluations is reached (N_{eval}) and a set of ARs is returned (RuleSet). Once all Maps are processed by the rule mining algorithm, the RuleSet from each Map is collected. Then, the global quality of the ARs of each RuleSet is evaluated using the entire dataset. To accomplish the above, antecedent support, consequent support and rule support are partially calculated for each Map, and they are aggregated to obtain the global evaluation that allow us to calculate the quality measures of the complete dataset. After that, all the ARs of each RuleSet are used to update an external set of rules which is termed GlobalRulePool, that store the non-dominated solutions found for the complete dataset considering the quality measures previously calculated. These quality measures will be the objective functions used by the sequential association rule algorithm. Subsequently, the redundant ARs of the GlobalRulePool are removed.

3.3 Computing the support of the model

Stage 2: Reducer Stage :

Input: supports a pair, key and value, in which key is the rule identity and value represents the antecedent support (supA), consequent support (supC) and rule support (supR) of such rule for each instance.

Output: (key, value) pair, where key is the id of each rule of RuleSet and value contains the global support of the antecedent, consequent and the rule considering all the instances.

Step 1: Rules $\leftarrow$ supports, reduce $\{(a,b) \Rightarrow$

Step 2: a.supA+=b.supA

Step 3: a.supC+=b.supC

Step 4: a.supR+=b.supR

Step 5: }

3.4 Global Rule Pool

The third part, denominated as GlobalRulePool, updates a global rule set in sequent that stores the non dominated solutions found in the entire dataset. To achieve that, the support obtained in the second phase is calculated to evaluate the quality of the ARs by the same measures used by the sequential algorithm.

Stage 3: Step by Step procedure

Input: (key, value) pair, where key is the id of each rule of RuleSet and value contains the antecedent global support, consequent and the rule considering all the instances.

Output: GlobalRulePool the external set of rules that contains the non-dominated solutions for the entire dataset.

Step 1: for RuleRS $\in$ RuleSet do

Step 2: if RuleRS satisfies Minimum Thresholds then

Step 3: for Rule GP $\in$ GlobalRulePool do

Step 4: if ruleRS dominates rule GB then

Step 5: remove rule GB from GlobalRulePool

Step 6: add ruleRS to Global RulePool

Step 7: end if

Step 8: if ruleRS and ruleGB are non-dominated solutions then

Step 9: add ruleRS to GlobalRulePool

Step 10: end if

Step 11: end for

Step 12: end if

Step 13: end for

Step 14: GlobalRulePool $\leftarrow$ remove Redundant (GlobalRulePool)

The fourth phase builds a new rule set of GlobalRulePool following a MapReduce scheme.

Stage 4: The fourth stage builds a new rule set of a GlobalRulePool by following a MapReduce scheme

Input: data that represents the dataset, each row is an Array of values.

Output: newData a new dataset with all uncovered instances.

Step 1: newData $\leftarrow$ data.filter (instance $\Rightarrow$

Step 2: keep $\leftarrow$ TRUE
Step 3: for Rule $\in$ GlobalRulePool do
Step 4: if instance is satisfied by Rule then
Step 5: keep $\leftarrow$ FALSE
Step 6: end if
Step 7: end for
Step 8: keep
Step 9: }

Brute-Force Pattern Matching Algorithm

- The brute force algorithm consists in checking, if the occurrence of the pattern starts at all positions or not. The positions in the input is (left to right) between 0 and $n - m$.
- The pattern is shifted towards right by one position after every attempt by the algorithm.
- It is considered as pre-computation is not done on the keyword in naive based algorithm
- Only a small constant amount of information with regard to its present position is kept and that is advantageous if the available memory space is to be considered.

Input:

$X \leftarrow$ m bytes array represents keyword
 $m \leftarrow$ keyword length integer
 $y \leftarrow$ n bytes array represents text input
 $n \leftarrow$ text length represents integer
Step 1: every character in y, for $j = 0 \rightarrow n-m$ do
Step 2: $i \leftarrow 0$
Step 3: while $i < m$ and $x[i] = y[i+j]$ do
Step 4: $i \leftarrow i+1$, $i =$ count of matching characters at and after y [j]
Step 5: end while
Step 6: if $i \geq m$ then
Step 7: output j
Step 8: end if
Step 9: end for
Step 10: end procedure

4. Result and Discussion

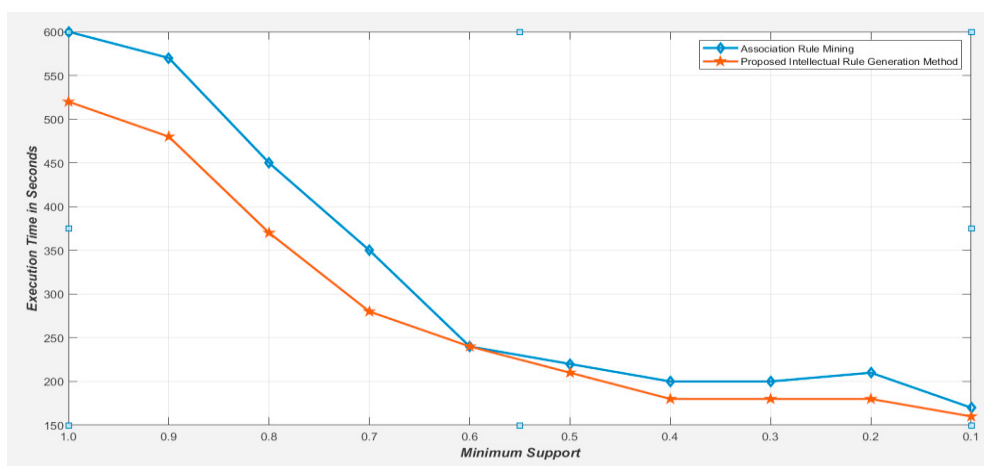
For the unknown attack, the rules structure is set by Rule Structure Generation and Pattern Matching algorithm generated by using Association Rule Mining in the Map Reduce Framework. The proposed Intellectual Rule Generation Method was evaluated on local standalone cluster installed in local machine. Java and MapReduce 2.0 (YARN) library are utilized for implementation of algorithms. The execution time of the Association Rule Mining (ARM) and proposed Intellectual Rule Generation Method (IRGM) (Table 1) with various minimum support values for given original dataset is represented in Fig 1 and Fig 2 respectively. The original dataset is pre-processed and classified by the previously proposed Precocious Classification framework [R]. The proposed framework is significant in that the original dataset is reduced and it is named as optimal dataset since it gives more accuracy and reduced error rates than the existing methods (Table 2; Fig 3). It is accomplished in two stages. In the first stage, an efficacious Rule structure is generated using Intellectual method. Pattern matching algorithm in Brute Force algorithm is utilized in the proposed frame work in the second stage. Notably intrusions are rare after proven analysis to prevent future attacks. The present HIDS proves to be a powerful design for understanding of erstwhile attacks and to ensure effective methods to defeat subsequent threats.

Table 1. Execution time of the proposed ARM and IRGM with varying minimum support for original dataset and optimal dataset

Minimum Support	Execution Time (in secs)			
	Original Data Set		Optimal Data Set	
	ARM	IRGM	ARM	IRGM
0.9	570	480	410	325
0.8	450	370	320	260
0.7	350	280	210	200
0.6	240	240	200	180
0.5	220	210	200	170
0.4	200	180	200	165
0.3	200	180	210	160
0.2	210	180	180	140
0.1	170	160	150	120

Table 2. Time elapse (in secs) of all MR phases with minimum support of 0.1

Number of Phases	Proposed Algorithm			
	Original Data Set		Optimal Data Set	
Phase 1	16	17	Phase 1	16
Phase 3	24	19	Phase 3	24
Phase 4	32	40	Phase 4	32
Phase 5	48	40	Phase 5	48
Phase 6	70	90	Phase 6	70
Phase 7	91	82	Phase 7	91
Phase 8	83	54	Phase 8	83
Phase 9	51	74	Phase 9	51
Phase 10	34	22	Phase 10	34
Total	467	458	Total	467
Actual	703	592	Actual	703

**Fig 2: Graphical Representation of the Execution time (in secs) with varying support value for ARM and proposed IRGM for Original Dataset**

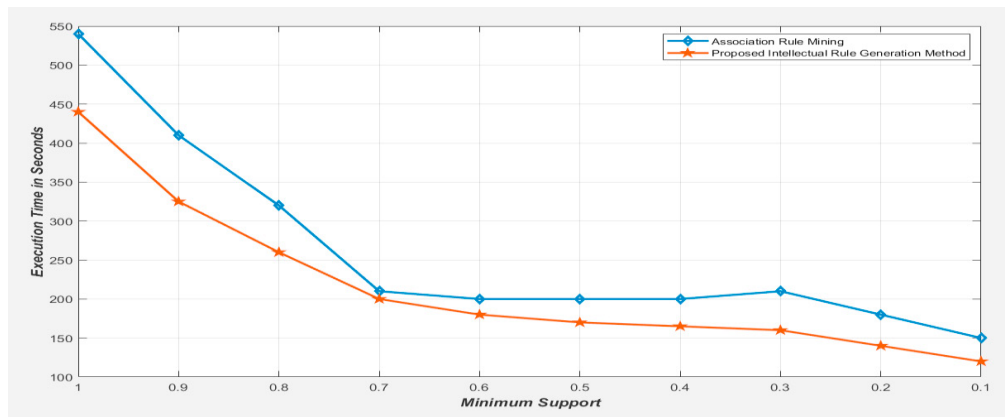


Fig 3: Graphical Representation of the Execution time (in seconds) with varying support value for ARM and proposed IRGM for Optimal Dataset

5. Conclusion

Intrusion detection systems (IDS) are mandatory for all types of networks to protect them from the intruders. The IDS have to process millions of network packets with many features, which delay the detection of anomalies. In order to protect data and information from various types of attack a novel intrusion detection system is required. The proposed host-based IDS proves to be a powerful mechanism for understanding of erstwhile intrusion and determining effective methods to defeat their subsequent threats. An Efficacious Intellectual Framework for Host based Intrusion Detection is proposed with Rule Structure Generation and Pattern Matching algorithm. The proposed Efficacious Intellectual Framework for Host based IDS used to better the quality of the Association Rule Mining. The quality of the rules is ensured by MapReduce framework. The four stages in the framework define the global support rules for host based IDS.

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